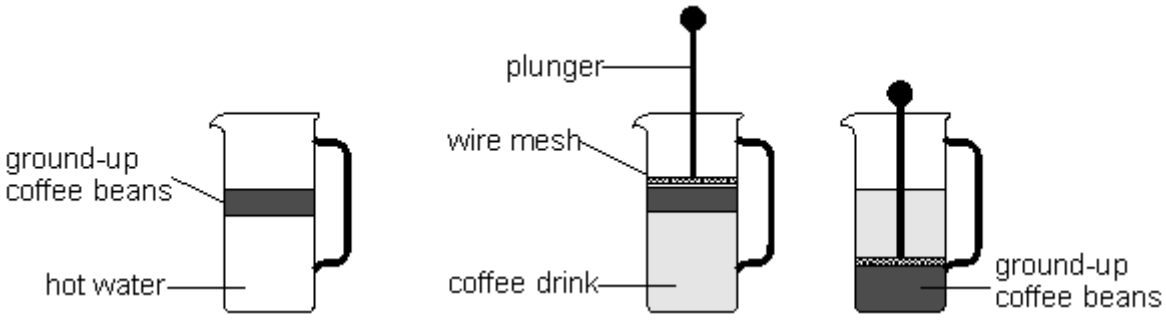
Q1. Russell put ground-up coffee beans in a coffee maker and added hot water.



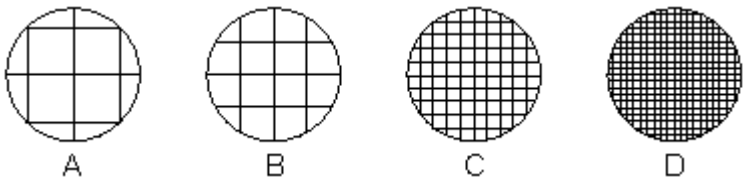
He pushed the plunger down.
This separated the coffee drink from the ground-up coffee beans.

(a) How could Russell see that some coffee had dissolved in the water?

.....

1 mark

(b) The end of the plunger is a circle of wire mesh.

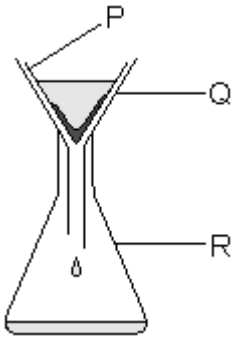


(i) Which mesh would be best to separate the coffee drink from all the ground-up coffee beans? Write the letter.

.....

1 mark

(ii) This method of making coffee uses a type of filter.
The apparatus used for filtration in a school laboratory is drawn below.

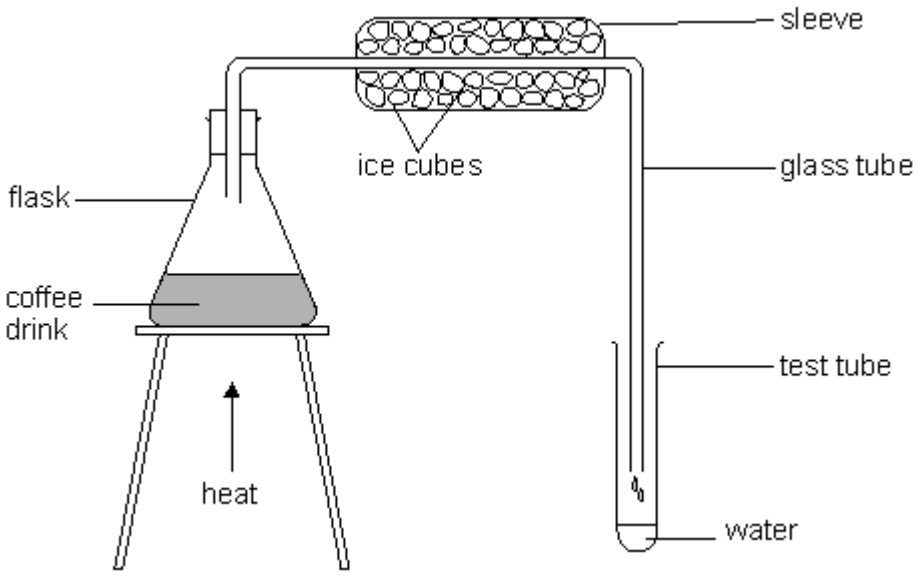


Which part of the apparatus above works in the same way as the wire mesh? Write the letter.

.....

1 mark

(c) Russell wanted to separate the water from the coffee drink.
He set up the apparatus shown below.



(i) Why did Russell put ice cubes around the glass tube?

.....

1 mark

(ii) Choose words from the box below to fill the gaps in the following sentences.

an acid	**a gas**	**a liquid**	**a solid**
condensation	**crystallisation**	**evaporation**	**filtration**

Russell heats the water. Water in the drink changes from

..... into ..... .

This change of state is called ..... .

Water vapour changes into liquid. This change of state is called

..... .

4 marks
maximum 8 marks

Q1.

(a) any **one** from

- the colour had changed
- the water was brown **or** darker **or** grey

accept 'the colour'
*do **not** accept 'the coffee disappeared'*

1 (L3)

(b) (i)
- D

1 (L3)

(ii)
- P

accept 'the filter paper'
'the filter' is insufficient
accept 'the paper'

1 (L3)

(c) (i) any **one** from

- to cool the water vapour
  *accept 'to cool the steam **or** gas'*
- to condense the water vapour
  *accept 'to condense the steam **or** gas'*
- change gas **or** vapour back to water
  *accept 'to cool it **or** the glass tube'*
  *do **not** accept references to liquid coffee condensing*

1 (L3)

(ii)
- *from a* liquid *into* a gas

2 (L4)

- evaporation

1 (L4)

- condensation

1 (L4)

award one mark for each correct word in the correct place

[8]